

The Service — What Sets the House's Capacity

Your most repeated error on this topic lives here. Three separate misses, all the same idea.

A house's electrical service capacity is set by the incoming conductor, not by the label on the panel. A system carries only what its most limited component allows, and that component is the wire coming in. A 200-amp panel fed by 100-amp conductors is a 100-amp service. The panel rating states only what the box is permitted to accept.

YOU HAVE MISSED THIS 3 TIMES

What can determine the home's electrical service capacity?

The size of the service drop wires.

YOU HAVE MISSED THIS 3 TIMES

The primary criterion for inlet service capacity is the:

Size of the inlet service wire

Those are the same fact asked two ways, and between them you have missed it six times. Whatever the wording — service drop, inlet service wire, service entrance conductor — the answer is the incoming wire.

Under-rated is dangerous, over-rated is merely oversized

YOU HAVE MISSED THIS 2 TIMES

A house has service entrance conductors designed for 100 amps and a main disconnect rated at 100 amps. Which condition is unsafe?

The service box is rated at 75 amps.

Read this class of question by looking for the component rated BELOW the rest. A 125-amp box on that same service would be oversized and harmless. A 75-amp box is asked to pass 100 amps, and that is the unsafe condition.

Inside the box

Line side	In the service box, the ungrounded service entrance wires connect to the line side of the main switch.
Six-throw rule	All power must be killable in six or fewer hand movements. A single main disconnect is not required so long as six throws will do it.
Working space	NEC 110.26 at a dwelling panel: depth 36 in, width 30 in, height 6 ft 6 in.

The 30 inches in your old notes is a width, not a depth

Your Number Families file lists "working space depth 30 in", which contradicts the 36 in it lists elsewhere. 36 is the depth, 30 is the width. The 30-inch depth that genuinely exists belongs to the gas furnace working space — same number, different appliance.

The numbers here

Panel working clearance — depth	36 in / 3 ft
Panel working clearance — width	30 in
Panel working clearance — height	6 ft 6 in
Minimum electric service	100 amp
Service size that indicates electric heat	200 amp

Say it back

Cover the page. Answer each one out loud.

1. What sets a house's service capacity, and what does the panel label tell you?
2. A 200-amp panel fed by 100-amp conductors is what size service?
3. 100-amp conductors, 100-amp main, 75-amp box. What is unsafe, and why?
4. State the three panel working-space dimensions and which is which.
5. How many hand movements may it take to kill all power?
6. What service size suggests the house has electric heat?